module-11-multiplication-near-base - Darshana (60-min workshop)

IKS Curriculum - senior band

Module 11 — Multiplication Near a Power of 10 (Darśana)

Sanskrit: *Nikhilam* · **Band:** Senior (9-12) · **Duration:** 60 min · **Path:** Darśana (Workshop)

Fast multiplication when numbers are near 10, 100, or 1000.

> **Darśana note.** This workshop condenses Module 11’s full 10-day arc into one hour. For the deeper version — daily lesson plans, quizzes, assessments, slide decks — switch to **Adhyayana** (the Full Course path).

Opening hook · 5 min

Project 998 × 997 = ? *“Long multiplication takes ~60 seconds. Nikhilam takes ~5 seconds. Today: the method, the algebra, and when (NOT) to use it.”*

Core idea · 15 min

Nikhilam multiplication — the algebra.

Identity: $(B - a)(B - b) = B^2 - B(a + b) + ab = B[B - (a + b)] + ab = B[(B - a) - b] + ab$.

So: - Cross-subtract = $(B - a) - b$ OR $(B - b) - a$ (both give $B - a - b$) - Multiply = ab

Padding: the “multiply” part must have as many digits as the base has zeros (e.g., for $B = 100$, pad to 2 digits).

Algebraic derivation FOR THE STUDENT (don’t just give them the rule): $(100 - 3)(100 - 4) = 10000 - 400 - 300 + 12 = 10000 - 700 + 12 = 9300 + 12 = 9312$.

When it’s faster: both factors WITHIN 5-10% of the same base. **When it’s slower:** factors far from any base.

Demonstration · 10 min

Algebra-on-the-board demo + edge cases. Derive identity. Then demo: 999 × 998 (deficits 1, 2 → 997 | 002 → 997002). Then show a mixed case: 102 × 98 (excess + deficit → cross-subtract → 100, multiply 2 × 2 = 4 → answer 9996, but pad: 102 × 98 = $100^2 - 4$ = 9996). Discuss.

Source moment · 5 min

Quoted reference. Tirthaji, *Vedic Mathematics* (1965), Sūtra 2 application to multiplication:

> “When two numbers are both close to the same power of 10, use the deficits (or excesses) to compute the product.”

Sūtra 2 (*Nikhilam*) is used for BOTH subtraction (Module 9) and multiplication (this module). Same algebra, two applications.

Hands-on activity · 15 min

Decision-tree poster. Each student designs a one-page flowchart: *“Given a multiplication problem, which method do I use? Long, ×11, Nikhilam, near-base Nikhilam, or other?”* Defend the boundaries.

Reflection + take-home · 10 min

“This module taught a special-case mental method. What’s the right balance in a curriculum between general algorithms and special-case tricks?”

Go deeper into Adhyayana

The 10 days you’re skipping if you only do this workshop:

- **Day 01** — Finger-trick prelude
- **Day 02** — 97 × 96 — two below base
- **Day 03** — 103 × 104 — two above base
- **Day 04** — Mixed cases
- **Day 05** — 3-digit near 1000
- **Day 06** — Algebra proof
- **Day 07** — When to use
- **Day 08** — Speed sprints
- **Day 09** — Mixed practice — combining ×11 and Nikhilam
- **Day 10** — Final assessment

→ **Open the Adhyayana track for this module** to access all 10 days, the 4-audience PDFs, the slide deck, the quizzes, and the project rubric.